

Kurka's $c_0 \oplus X$ Borel Question: Verification, Corrections, and Sharpened Obstructions

Research note prepared from the supplied source paper and partial solution

Literature checked through 18 June 2026

Abstract

Kurka asked whether there is a separable Banach space X for which the isomorphism class of $c_0 \oplus X$ is Borel. I verify the main arguments in the supplied partial solution, correct one substantive point, and derive several stronger necessary conditions for any positive example. The key correction is that the proposed “paving entry point” is vacuous: Johnson–Lindenstrauss–Schechtman proved that a space determined by its pavings has nontrivial type, whereas every space containing c_0 has only type one. Thus no positive example can be determined by its pavings or by its finite-dimensional subspaces.

The main new structural consequence is that a Borel positive example Y would satisfy $Y/E \simeq Y$ for every copy $E \simeq c_0$ in Y , and every isomorphism between two copies of c_0 in Y would extend to an automorphism of Y . A sharpened form of Kurka's Bourgain–Delbaen reduction shows that such a Y must absorb, as complemented summands with complement again isomorphic to Y , a properly coanalytic set of finite-only parameters. That parameter set has unbounded Cantor–Bendixson rank below ω_1 . These conclusions significantly narrow the shape of a possible example, but they do not yield a contradiction. No complete proof or counterexample was located in the literature through the date above; in particular, the 2024 paper of Cúth–Doležal–Doucha–Kurka still asks whether the Gurarii space has a Borel isomorphism class.

1 The question and the outcome of the search

Let SB denote any standard Borel coding of separable Banach spaces, and let

$$[Y]_{\cong} = \{Z \in \text{SB} : Z \simeq Y\}.$$

Kurka proved that $[c_0]_{\cong}$ is complete analytic and asked whether there is a separable X such that $[c_0 \oplus X]_{\cong}$ is Borel [6]. His Theorem 1.2 gives a negative answer when X omits c_0 , when X has no infinite-dimensional reflexive subspace, or when X embeds into a space with an unconditional basis.

only. Theorem 1.1 is an improvement indeed, since it is basically a statement about the structure of the class of $\mathcal{L}_\infty$ -spaces.

Actually, it follows from our method of proving Theorem 1.1 that the isomorphism class of $c_0 \oplus X$ is not Borel in the case that X is not big in a sense (in fact, we do not know if there is a space X such that the isomorphism class of $c_0 \oplus X$ is Borel).

Theorem 1.2. *Let X be a separable Banach space that satisfies any of the following conditions:*

- *X does not contain an isomorphic copy of c_0 ,*
- *X does not contain an infinite-dimensional reflexive subspace,*
- *X is a subspace of a space with an unconditional basis.*

Then the class of all Banach spaces isomorphic to $c_0 \oplus X$ is not Borel. Furthermore, the class of all Banach spaces isomorphic to a subspace of $c_0 \oplus X$ is not Borel as well.

Let us remark that the second part of this result has been already

The literature search did not locate a later resolution of the general question. The strongest current checkpoint found is the published paper of Cúth, Doležal, Doucha, and Kurka: they explicitly state that they do not know whether the isomorphism class of the Gurarii space is Borel and formulate this as an open question [1, Question 2]. A 2026 paper of Smith on a nearby isomorphism-class problem continues to cite Kurka’s result as part of the current descriptive-set-theoretic landscape [7]. This is not a proof that no resolution exists, but it is consistent with the direct searches of journal, arXiv, and citation records.

Research outcome. The partial solution is mostly correct, but one claimed route is impossible. The present note supplies stronger necessary conditions and a precise descriptive-set-theoretic bottleneck. It does not claim a complete solution of Kurka’s question.

Run status. In the `fa_banach_001` registry this packet should be treated as a stronger partial result: it sharpens the previous partial packet and corrects its paving-route interpretation, but it neither constructs a positive Borel example nor proves that none exists.

2 Audit of the supplied partial result

Claim in the packet	Verdict	Comment
Sobczyk–Kurka characterization	Correct	The absorption $X \simeq c_0 \oplus X$ follows immediately from Sobczyk’s theorem when X contains c_0 .
Quotient-persistence obstruction	Correct, and sharpened	In a Borel positive example, every quotient by a copy of c_0 is not merely “large”: it is isomorphic to the original space.
Unconditional-embedding exclusion	Correct	It is a direct application of Kurka’s Theorem 1.2(iii) after splitting off c_0 .
No F_σ positive example; Gurarii is the only possible G_δ example	Correct in the CDDK codings	The assertions F_σ , G_δ , and meager depend on the chosen Polish topology; plain Borelness is the coding-invariant standard-Borel issue.
Unique nonmeager isomorphism class	Correct in the CDDK codings	It follows from the dense G_δ isometry class of the Gurarii space and the Baire property of analytic isomorphism classes.
Paving entry point	Formally vacuous; substantively misleading	A space determined by its pavings has type $2 - \varepsilon$ for every $\varepsilon > 0$, so it cannot contain c_0 . This should be replaced by a paving exclusion theorem.
Collapse-set obstruction	Correct	The Lusin-separation argument is sound. It can be strengthened from “nonanalytic excess” to a properly coanalytic, Cantor–Bendixson-rank-unbounded excess.

Two presentational corrections are also useful. First, the relevant numbered results in the published version of Kurka’s paper are Proposition 4.5, Proposition 4.7, Lemma 5.1, Theorem 5.2, and the proof of Theorem 1.2. Second, one should say that the *parameter set* is nonanalytic; the map $\mathcal{M} \mapsto X_{\mathcal{M}}$ need not be injective up to isomorphism.

3 The basic Sobczyk–Kurka dichotomy

Theorem 3.1 (Sobczyk–Kurka dichotomy). *For every separable Banach space X ,*

$$[c_0 \oplus X]_{\cong} \text{ is Borel} \iff X \text{ contains } c_0 \text{ and } [X]_{\cong} \text{ is Borel.}$$

Equivalently, Kurka’s question asks whether any separable Banach space containing c_0 has a Borel isomorphism class.

Proof. If X contains no copy of c_0 , Kurka’s Theorem 1.2(i) says that $[c_0 \oplus X]_{\cong}$ is not Borel.

Suppose that $E \subseteq X$ and $E \simeq c_0$. Sobczyk’s theorem says that E is complemented in the separable space X , so $X = E \oplus Z$ for some closed Z . Since $c_0 \oplus c_0 \simeq c_0$,

$$c_0 \oplus X \simeq c_0 \oplus E \oplus Z \simeq c_0 \oplus c_0 \oplus Z \simeq c_0 \oplus Z \simeq X.$$

Thus $[c_0 \oplus X]_{\cong} = [X]_{\cong}$ whenever X contains c_0 . □

Corollary 3.2. *For a separable Banach space X , the following are equivalent:*

$$X \text{ contains } c_0, \quad X \text{ contains a complemented copy of } c_0, \quad X \simeq c_0 \oplus X.$$

4 A stronger quotient theorem and restricted automorphy

The packet records only that quotients by copies of c_0 must avoid the three obstructions in Kurka's theorem. Borelness forces much more.

Theorem 4.1 (Quotient self-similarity). *Let Y be separable, suppose that Y contains c_0 , and assume that $[Y]_{\cong}$ is Borel. Then, for every subspace $E \subseteq Y$ with $E \simeq c_0$,*

$$Y/E \simeq Y.$$

Equivalently, every complement of every copy of c_0 in Y is isomorphic to Y .

Proof. By Sobczyk's theorem, write $Y = E \oplus Z$, where $Z \simeq Y/E$. Hence $Y \simeq c_0 \oplus Z$. If Z contained no c_0 , Kurka's Theorem 1.2(i) would imply that $[c_0 \oplus Z]_{\cong} = [Y]_{\cong}$ is not Borel, a contradiction. Therefore Z contains c_0 . Applying Corollary 3.2 to Z ,

$$Z \simeq c_0 \oplus Z \simeq Y.$$

□

Corollary 4.2 (Extension of isomorphisms between c_0 -copies). *Under the hypotheses of Theorem 4.1, let $E, F \subseteq Y$ be copies of c_0 . Every isomorphism $u : E \rightarrow F$ extends to an automorphism of Y . In particular, $\text{Aut}(Y)$ acts transitively on the copies of c_0 in Y .*

Proof. Choose complements $Y = E \oplus Z = F \oplus W$. By Theorem 4.1, $Z \simeq Y \simeq W$; let $v : Z \rightarrow W$ be an isomorphism. Then

$$U(e + z) = u(e) + v(z) \quad (e \in E, z \in Z)$$

defines an automorphism of Y extending u .

□

Remark 4.3. Corollary 4.2 is a restricted automorphy property: only subspaces isomorphic to c_0 are involved. No theorem was found that makes this property contradictory in a general separable Banach space. Thus this sharpening is a useful test for candidates, but not by itself a negative solution.

Corollary 4.4 (Unconditional exclusion). *If a separable space Y contains c_0 and embeds into a Banach space with an unconditional basis, then $[Y]_{\cong}$ is not Borel. Consequently, a positive example can have neither an unconditional basis nor an embedding into c_0 . In particular, no space $(\bigoplus_n E_n)_{c_0}$, with all E_n finite-dimensional, can be a positive example.*

Proof. Split $Y \simeq c_0 \oplus Z$ by Sobczyk. The complement Z embeds into the same unconditional-basis space as Y , so Kurka's Theorem 1.2(iii) applied to Z gives the first assertion. For the last assertion, choose uniformly bounded embeddings $E_n \hookrightarrow \ell_{\infty}^{k_n}$; their c_0 -sum embeds into $(\bigoplus_n \ell_{\infty}^{k_n})_{c_0} \simeq c_0$. □

5 Correction: pavings are an exclusion, not an entry point

Theorem 5.1 (Johnson–Lindenstrauss–Schechtman). *If a separable Banach space is determined by its pavings, then for every $\varepsilon > 0$ it has type $2 - \varepsilon$ and cotype $2 + \varepsilon$ [4, Proposition 7.4].*

Lemma 5.2. *Every Banach space containing an isomorphic copy of c_0 has no type $p > 1$.*

Proof. It is enough to prove this for c_0 , since type passes to subspaces and is invariant under isomorphism. For $n \in \mathbb{N}$, use coordinates indexed by $\{-1, 1\}^n$, and define $x_i \in \ell_\infty^{2^n} \subseteq c_0$ by

$$x_i(\sigma) = \sigma_i \quad (1 \leq i \leq n).$$

Then $\|x_i\| = 1$. For every choice of signs $\varepsilon_1, \dots, \varepsilon_n$, the coordinate $\sigma = (\varepsilon_1, \dots, \varepsilon_n)$ gives

$$\left\| \sum_{i=1}^n \varepsilon_i x_i \right\|_\infty = n.$$

A type- p inequality would therefore imply $n \leq Tn^{1/p}$ for all n , impossible when $p > 1$. $\square$

Corollary 5.3 (Paving exclusion). *No separable Banach space containing c_0 is determined by its pavings. Consequently, no such space is determined by its finite-dimensional subspaces.*

Proof. Theorem 5.1, with for example $\varepsilon = 1/2$, gives nontrivial type, contradicting Lemma 5.2. A space determined by its finite-dimensional subspaces is determined by its pavings [4, Definition 7.2 and the following observation]. $\square$

Thus the packet's conditional statement "a c_0 -containing space determined by its pavings would be a positive example" is logically true but has an impossible hypothesis. It should not be advertised as a construction route.

6 The CDDK topological barrier

This section concerns the particular Polish codings $\mathcal{B}$ and $\mathcal{P}_\infty$ of Cúth–Doležal–Doucha–Kurka. The Borel/non-Borel question is standard-Borel in nature, but the finer labels F_σ , G_δ , meager, and comeager refer to these topologies.

Theorem 6.1 (Low-complexity barrier). *Let Y be a separable infinite-dimensional Banach space containing c_0 .*

- (i) *In $\mathcal{B}$ and $\mathcal{P}_\infty$, $[Y]_\cong$ is not F_σ .*
- (ii) *If $[Y]_\cong$ is G_δ , then $Y \simeq \mathbb{G}$, the Gurarii space.*
- (iii) *If $Y \not\simeq \mathbb{G}$, then $[Y]_\cong$ is meager. The class $[\mathbb{G}]_\cong$ is comeager.*

Proof. Since Y contains c_0 , it is not isomorphic to ℓ_2 . CDDK prove that ℓ_2 is the unique separable infinite-dimensional space with an F_σ isomorphism class in either coding [1, Theorem 2.10], proving (i). They also prove that, with the possible exception of $\mathbb{G}$, no isomorphism class is G_δ [1, Proposition 2.11], proving (ii).

For (iii), every isomorphism class is analytic and therefore has the Baire property. If $[Y]_\cong$ is nonmeager, it is comeager in some nonempty open set. The isometry class of $\mathbb{G}$ is a dense G_δ set in both codings [1, Theorem 1.10], so it meets $[Y]_\cong$. Hence $Y \simeq \mathbb{G}$. Conversely, $[\mathbb{G}]_\cong$ contains that dense G_δ isometry class and is therefore comeager. $\square$

Corollary 6.2 (Route dichotomy). *In the CDDK codings, a positive answer to Kurka's question must arise in one of exactly two ways:*

- (a) *$[\mathbb{G}]_\cong$ is Borel; or*
- (b) *there is a c_0 -containing space with a meager Borel isomorphism class which is not F_σ .*

A nonmeager Borel positive example exists if and only if $[\mathbb{G}]_\cong$ is Borel.

7 A sharpened absorption-set obstruction

We now use Kurka's parameterized Bourgain–Delbaen construction. Let

$$\mathcal{H} = \{\mathcal{M} \in \mathcal{K}(\mathcal{P}(\mathbb{N})) : \mathcal{M} \text{ contains an infinite set}\}.$$

Hurewicz's theorem says that $\mathcal{H}$ is complete analytic and non-Borel. Kurka constructs a Borel assignment $\mathcal{M} \mapsto X_{\mathcal{M}}$ with the following properties [6, Propositions 4.5 and 4.7, Lemma 5.1]:

- (K1) if $\mathcal{M} \in \mathcal{H}$, then $X_{\mathcal{M}} \simeq c_0$;
- (K2) if $\mathcal{M} \notin \mathcal{H}$, then $X_{\mathcal{M}}$ contains no copy of c_0 ;
- (K3) there is a Borel set $\mathcal{B}_0 \supseteq \mathcal{H}$ such that $X_{\mathcal{M}}^*$ is separable for $\mathcal{M} \in \mathcal{B}_0$;
- (K4) for $\mathcal{M} \in \mathcal{B}_0 \setminus \mathcal{H}$, every infinite-dimensional subspace of $X_{\mathcal{M}}$ contains an infinite-dimensional reflexive subspace.

For fixed Y , Kurka's coding also gives a Borel map T_Y with $T_Y(\mathcal{M}) \simeq X_{\mathcal{M}} \oplus Y$.

Definition 7.1. For a fixed separable space Y , define its Kurka absorption set by

$$\mathfrak{A}_Y = \{\mathcal{M} \in \mathcal{B}_0 : X_{\mathcal{M}} \oplus Y \simeq Y\}.$$

Theorem 7.2 (Properly coanalytic absorption excess). *Suppose that Y contains c_0 and $[Y]_{\cong}$ is Borel. Then:*

- (i) $\mathfrak{A}_Y$ is Borel and $\mathcal{H} \subseteq \mathfrak{A}_Y$;
- (ii) $\mathfrak{A}_Y \setminus \mathcal{H}$ is coanalytic but not analytic;
- (iii) for every $\mathcal{M} \in \mathfrak{A}_Y \setminus \mathcal{H}$, the space Y contains a complemented copy of $X_{\mathcal{M}}$, with complement isomorphic to Y ;
- (iv) each such $X_{\mathcal{M}}$ contains no c_0 , has separable dual, and is reflexively saturated.

Proof. Since T_Y is Borel and $[Y]_{\cong}$ is Borel,

$$\mathfrak{A}_Y = \mathcal{B}_0 \cap T_Y^{-1}([Y]_{\cong})$$

is Borel. If $\mathcal{M} \in \mathcal{H}$, then $X_{\mathcal{M}} \simeq c_0$, while $Y \simeq c_0 \oplus Y$ by Corollary 3.2; hence $\mathcal{M} \in \mathfrak{A}_Y$.

The set $\mathfrak{A}_Y \setminus \mathcal{H} = \mathfrak{A}_Y \cap \mathcal{H}^c$ is coanalytic. Suppose it were analytic. The disjoint analytic sets $\mathcal{H}$ and $\mathfrak{A}_Y \setminus \mathcal{H}$ could then be separated by a Borel set D , with $\mathcal{H} \subseteq D$ and $D \cap (\mathfrak{A}_Y \setminus \mathcal{H}) = \emptyset$. Since $\mathcal{H} \subseteq \mathfrak{A}_Y$, this would give

$$\mathcal{H} = D \cap \mathfrak{A}_Y,$$

contradicting the non-Borelness of $\mathcal{H}$.

For $\mathcal{M} \in \mathfrak{A}_Y \setminus \mathcal{H}$, transport the two coordinate projections through an isomorphism $X_{\mathcal{M}} \oplus Y \rightarrow Y$. This yields a complemented copy of $X_{\mathcal{M}}$ in Y , whose complementary summand is isomorphic to Y . The last statement follows from (K2)–(K4). $\square$

Corollary 7.3 (Finite absorption). *Under the hypotheses of Theorem 7.2, suppose that*

$$\mathcal{M}_1, \dots, \mathcal{M}_n \in \mathfrak{A}_Y \setminus \mathcal{H}.$$

Then

$$Y \simeq \left(\bigoplus_{j=1}^n X_{\mathcal{M}_j} \right) \oplus Y.$$

Proof. Iterate the identities $Y \simeq X_{\mathcal{M}_j} \oplus Y$. $\square$

8 Unbounded Cantor–Bendixson parameter rank

If $\mathcal{M} \notin \mathcal{H}$, every member of $\mathcal{M}$ is finite. Therefore $\mathcal{M}$ is a countable compact, hence scattered, subset of the Cantor space $2^{\mathbb{N}}$. Let $\rho_{\text{CB}}(\mathcal{M})$ denote its Cantor–Bendixson rank. The Cantor–Bendixson rank is a standard coanalytic rank on the space of countable compact subsets; in particular, for each $\alpha < \omega_1$,

$$R_\alpha = \{\mathcal{M} \in \mathcal{H}^c : \rho_{\text{CB}}(\mathcal{M}) \leq \alpha\}$$

is Borel [5, Section 34 and the boundedness theorem for coanalytic ranks].

Theorem 8.1 (Rank-unbounded absorption). *If Y is a Borel positive example, then*

$$\sup\{\rho_{\text{CB}}(\mathcal{M}) : \mathcal{M} \in \mathfrak{A}_Y \setminus \mathcal{H}\} = \omega_1.$$

Equivalently, for every countable ordinal α , there is a finite-only parameter $\mathcal{M} \in \mathfrak{A}_Y$ with $\rho_{\text{CB}}(\mathcal{M}) > \alpha$.

Proof. If the ranks on $\mathfrak{A}_Y \setminus \mathcal{H}$ were bounded by some $\alpha < \omega_1$, then

$$\mathfrak{A}_Y \setminus \mathcal{H} = \mathfrak{A}_Y \cap R_\alpha$$

would be Borel. This contradicts Theorem 7.2(ii). $\square$

Remark 8.2. Theorem 8.1 concerns the complexity of the parameter set. It does not assert that the spaces $X_{\mathcal{M}}$ obtained at different ranks are pairwise nonisomorphic. Establishing an isomorphic invariant of $X_{\mathcal{M}}$ that grows with $\rho_{\text{CB}}(\mathcal{M})$ would be a possible route to a contradiction, but no such invariant was found in Kurka’s family. In particular, Kurka may choose $\mathcal{B}_0$ so that the relevant spaces have uniformly small Szlenk index, which blocks the most immediate Szlenk-rank argument.

9 Attempts toward a full resolution

The following routes were examined.

1. **Find a maximal c_0 -summand.** One might hope that every separable Y containing c_0 has a copy $E \simeq c_0$ for which Y/E omits c_0 , contradicting Theorem 4.1. No such general theorem holds by a straightforward Zorn or recursive argument: successive Sobczyk splittings do not provide uniformly controlled projections at limit stages.
2. **Turn quotient self-similarity into an unconditional decomposition.** The identity $Y \simeq c_0 \oplus Y$ can be iterated, but it already holds for every separable space containing c_0 . The resulting projection norms need not be uniformly bounded, so the iteration does not force an unconditional decomposition or an embedding into a space with an unconditional basis.
3. **Use restricted automorphy.** Corollary 4.2 is strong, but it concerns only copies of c_0 . Existing automorphic-space obstructions do not directly rule out this restricted property.
4. **Use pavings or finite-dimensional determination.** This route closes completely, but negatively: Corollary 5.3 shows that no c_0 -containing space can have either property.

5. **Bound the complexity of Kurka’s absorbed summands.** Theorem 8.1 shows that any positive example must absorb parameters of arbitrarily high Cantor–Bendixson rank. A full negative solution would follow from a theorem bounding that rank for summands complementably absorbed by a fixed Y . No such bound is presently available, and the rank is a rank of parameters rather than a known isomorphic invariant of the resulting spaces.
6. **Force analyticity of the finite-only collapse set.** This is the cleanest exact bottleneck. For a fixed Y , the relation $X_{\mathcal{M}} \oplus Y \simeq Y$ is analytic in $\mathcal{M}$; under the Borelness assumption it becomes Borel. A general structural theorem making its finite-only part analytic would contradict Theorem 7.2. No such theorem was found.

10 Conclusion

The supplied partial result contains a sound central reduction: Kurka’s question is exactly the search for a Borel isomorphism class represented by a separable space containing c_0 . Its quotient, unconditional, CDDK, and Lusin-separation arguments survive verification. The paving claim requires a substantive correction: pavings exclude, rather than produce, candidates.

Any positive example Y would have to satisfy all of the following at once:

- (1) every quotient Y/E , for $E \simeq c_0$, is isomorphic to Y ;
- (2) every isomorphism between copies of c_0 extends to an automorphism of Y ;
- (3) Y embeds into no space with an unconditional basis and is not determined by pavings;
- (4) in the CDDK codings, either $Y \simeq \mathbb{G}$, or $[Y]_{\cong}$ is meager, Borel, and non- F_{σ} ;
- (5) Y complementably absorbs a properly coanalytic, rank-unbounded set of Kurka Bourgain–Delbaen summands, with complement again isomorphic to Y .

These conditions make a positive example highly rigid, but no contradiction or explicit example is presently obtained.

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